

A Unified Robust Low-Rank Framework for Time Synchronization in Multi-Drone Networks

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Abstract—This paper addresses the challenge of clock synchronization in multi-drone networks operating within dynamic, interference-prone low-altitude airspace. In such environments, interference and packet loss frequently lead to clock misalignment and drift accumulation. We model the measured clock data as a decomposition of low-rank, sparse, and noise components. The study evaluates five synchronization strategies: a classical Baseline, standard low-rank recovery methods (Fill+SVD, Soft Impute), robust decomposition (GoDec), and our proposed Graph-Regularized Robust Low-Rank (GR-RLR) model. The GR-RLR model uniquely integrates spatial graph Laplacian smoothness with temporal-fused regularization to fully exploit spatio-temporal priors. Simulation results validate the framework, demonstrating that GR-RLR offers superior robustness and stable synchronization accuracy under heavy jamming and packet-loss conditions, significantly outperforming conventional methods in suppressing impulsive errors.

I. INTRODUCTION

Accurate clock synchronization is essential in multi-drone and autonomous swarm systems to maintain communication, coherent sensing, and control. Cooperative sensing, such as localization and formation control, degrades as the number of agents grows due to wireless impairments, including interference, packet loss, and signal fading, leading to asynchronous timestamps and drift accumulation. The reliability of the classical consensus and affine model [1]–[3] has been limited, especially when timestamp data is missing or corrupted.

These conventional methods typically struggle in harsh low-altitude airspace because they are not designed to handle the non-Gaussian, impulsive errors caused by jamming or the large blocks of missing data from communication loss. However, the collective clock data from a multi-drone swarm exhibits two key structural properties that can be exploited: (i) the temporal evolution of all drone clocks is highly correlated (as they are driven by similar physical drift processes), which manifests as a **low-rank structure** in the observation matrix; and (ii) severe interference events and packet losses are sporadic and affect only a subset of drones at any given time, resulting in **sparse outliers** or missing entries.

This insight allows us to reframe the synchronization challenge as a robust low-rank matrix recovery problem, rather

than one of simple filtering. This reframing naturally leads to recent advancements in low-rank modeling [4]–[7] and graph-based optimization [8], [9], which have demonstrated promising performance in recovering signals from incomplete observations and noisy environments.

This work presents a unified approach to clock synchronization for multiple drones operating in these challenging conditions. This framework models the behavior of the entire swarm and allows errors and missing data to be corrected jointly. Five synchronization strategies have been used in this work, including baseline detrending, Fill+SVD, and Soft-Impute, GoDecomposition, and our proposed graph-regularized robust low-rank (GR-RLR) model. The GR-RLR brings two ideas together: inter-drone relationships via spatial graph regularization and robust loss functions to preserve temporal consistency. This hybrid design can enable recovery of corrupted or missing data even in the presence of extreme interference. This framework establishes a link between communication topology, data reliability, and synchronization performance, providing new insights into distributed coordination. The work advances synchronization theory toward scalable, interference-resilient operation for large autonomous aerial networks.

II. RELATED WORK

The recent evolution in multi-drone synchronization has transcended classical consensus and IEEE 1588-based solutions, moving toward approaches that directly address packet loss, jamming, and non-homogeneous channel environments. To support UAV ad hoc networks, a synchronization scheme [10] was proposed, demonstrating that basic alignment can be achieved despite mobility and link variability. Still, when adversarial interference is present, performance suffers. Precision Time Protocol (PTP) IEEE 1588-2019 is a standards-based framework that provides packet-based synchronization across heterogeneous systems [11].

Recently, low-rank recovery and matrix completion frameworks have emerged as vital tools for modeling correlated temporal behavior. Low-rank structure modeling, which can reconstruct a signal from noisy, sparse measurements, is often used in distributed estimation [12]. Online matrix completion and dynamic principal component pursuit further improved adaptability to time-varying conditions [13]. However, these methods usually assume independent observations and overlook inter-agent graph correlations.

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Graph signal processing (GSP) and Laplacian regularization have been employed in control and communication systems to address spatial interdependence. Graph-based smoothness is used to capture internode dependencies [14] and can be applied to structured data recovery using graph Laplacian learning [14]. Incorporating such priors with robust norms (like Huber or l_1 based penalties) has been used to improve resilience in multi-agent sensing systems.

The literature reviewed above, however, reveals key limitations in existing approaches. First, classical consensus [1], [2] and affine models [3], [10] are ill-equipped to handle the sparse, high-magnitude corruptions common in aerial networks. Second, while standard low-rank recovery methods [12], [13] can address sparse outliers, they typically ignore the underlying network topology (i.e., the spatial correlation). Finally, while GSP-based methods [8], [9], [14] effectively leverage this spatial structure, they often lack explicit mechanisms for robustly handling large, sparse corruptions and missing data simultaneously. Therefore, a *unified framework* that cohesively integrates robust low-rank decomposition with spatio-temporal graph priors is still lacking. This paper aims to fill this gap.

III. SYSTEM MODELING

Consider an autonomous network of drones D forming a cooperative synchronization cluster, in which the reference (true) clock is denoted by $x^*(t)$ and the internal clock of drone i is expressed as

$$y_i(t) = x^*(t) + b_i + d_i t + \varepsilon_i(t), \quad (1)$$

where b_i represents the static offset, d_i is the linear drift, and $\varepsilon_i(t)$ denotes the stochastic error term. At a uniform sampling interval $t_n = n\Delta t$ for $n = 0, \dots, N-1$, we obtain the discrete clock matrix:

$$\mathbf{Y} = [y_i[n]] \in \mathbb{R}^{D \times N}. \quad (2)$$

Environmental effects (e.g., interference, signal attenuation) can prevent correct timestamp reception, yielding corrupted or missing entries. The observed matrix is thus defined as $\mathbf{M} = \mathbf{W} \odot \mathbf{Y}$, where $\mathbf{W} \in \{0, 1\}^{D \times N}$ is an observation mask and $\odot$ denotes the elementwise product.

IV. DECOMPOSITION FRAMEWORK

To restore synchronization timing, we adopt a decomposition framework that separates the corrupted observation matrix $\mathbf{M}$ into its constituent parts. As shown in (3), the model is designed to isolate the underlying low-rank structure (the clean signal) from sparse corruptions and random noise:

$$\mathbf{M} = \mathbf{L} + \mathbf{S} + \mathbf{N}. \quad (3)$$

In (3), $\mathbf{L}$ captures low-rank trends due to correlated clock drifts, whereas $\mathbf{S}$ models sparse or impulsive errors caused by environmental effects, and $\mathbf{N}$ denotes small random noise. Each method for estimating $\mathbf{L}$ and $\mathbf{S}$ using mathematical modeling is described below.

This work's contribution does not lie in proposing a new decomposition method but in the principled integration and

task-specific adaptation of multiple robustness mechanisms for clock synchronization in UAV swarms under adversarial and heterogeneous sensing conditions. To the best of our knowledge, this is the first framework that jointly incorporates (i) robust data fidelity via Huber loss to handle impulsive timing errors, (ii) sparse corruption explicitly modeling jammed or faulty drones, (iii) graph Laplacian regularization to enforce spatial consistency over a communication topology, and (iv) temporal fused regularization to preserve clock continuity across time, all within a unified optimization formulation tailored to multi-drone clock recovery. Existing methods typically address only subsets of these challenges in isolation.

We will now evaluate five strategies for solving this decomposition problem, presented in order of increasing model complexity. We begin with a classical **Baseline** method, which serves as a performance reference and does not exploit the data's low-rank structure. We then progressively evaluate four methods based on low-rank matrix recovery: **Fill+SVD**, **Soft Impute**, **Go-Decomposition**, and our proposed **Graph-Regularized Robust Low-Rank (GR-RLR)** model. This comparison will demonstrate the advantages of incorporating robust statistics and spatio-temporal priors, as is done in the GR-RLR framework.

A. Base Line method

Each drone's clock deviation from the reference clock is approximated by an affine model as

$$y_i[n] = \alpha_i + b_i t_n + \varepsilon_i[n], \quad (4)$$

where the parameters α_i and b_i are estimated through robust regression or least squares. The synchronized signal is reconstructed as

$$\hat{y}_i[n] = y_i[n] - (\alpha_i + b_i t_n). \quad (5)$$

This baseline method compensates for drift and offset but does not exploit inter-drone correlation.

B. Fill+ SVD

Firstly, missing measurements are filled/interpolated using temporal extrapolation and column-wise median correction, producing a complete matrix $\mathbf{M}$. Then, a low-rank matrix is extracted with singular value decomposition given by

$$\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \quad (6)$$

where $\mathbf{U} \in \mathbb{R}^{D \times r}$, $\mathbf{V} \in \mathbb{R}^{N \times r}$, and $\mathbf{\Sigma} = \text{diag}(s_1, \dots, s_r)$. The effective rank r is chosen to satisfy

$$\min_r \frac{\sum_{i=1}^r s_i^2}{\sum_{i=1}^{\min(D,N)} s_i^2} \geq \eta \quad (7)$$

where $\eta \in [0, 1]$ is an energy threshold. Residual noise is mitigated in (8) by column-wise shrinkage, leading to the synchronized estimation:

$$\hat{\mathbf{Y}}_{\text{Fil}} = \mathbf{Y} - (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T + \mathbf{S}_{\text{robust}}). \quad (8)$$

C. Soft Impute Method

The synchronization problem is formulated in (9) to address missing or corrupted timestamps caused by wireless interference.

$$\min_{\mathbf{L}} \frac{1}{2} \|\mathbf{W} \odot (\mathbf{M} - \mathbf{L})\|_F^2 + \lambda \|\mathbf{L}\|_* \quad (9)$$

Where $\|\cdot\|_F$ is the Frobenius norm, $\|\mathbf{L}\|_*$ is the nuclear norm, $\mathbf{M}$ is the matrix of observed drone timestamps, and $\mathbf{W}$ is the binary reception mask. Optimal reconstruction is achieved in (10) by iterative singular value thresholding:

$$\mathbf{L}^{(k+1)} = \mathcal{D}_\lambda(\mathbf{W} \odot \mathbf{M} + (\mathbf{1} - \mathbf{W}) \odot \mathbf{L}^{(k)}). \quad (10)$$

D. Go-Decomposition

The Go-Decomposition model observes the drone timestamp matrix as the sum of a structured low-rank component and a sparse error term. That is

$$\mathbf{M} = \mathbf{L} + \mathbf{S} \quad \text{s.t. } \text{rank}(\mathbf{L}) \leq r, \|\mathbf{S}\|_0 \leq K, \quad (11)$$

where $\mathbf{L}$ denotes coherent clock behavior across drones, while $\mathbf{S}$ captures impulsive errors or packet anomalies and $\|\mathbf{S}\|_0$ counts non-zero elements. This method alternates between two projections: Low-rank: $\mathbf{L} \leftarrow \mathcal{P}_r(\mathbf{M} - \mathbf{S})$, where $\mathcal{P}_r$ represents the best rank- r approximation. Sparse: retain the K largest magnitude residuals $\mathbf{S} \leftarrow \mathcal{T}_K(\mathbf{M} - \mathbf{L})$. After convergence, the synchronization estimation is calculated as $\hat{\mathbf{Y}}_{\text{GoDec}} = \mathbf{Y} - (\mathbf{L} + \mathbf{S})$.

E. Graph Regularized Robust Low-Rank (GR-RLR) Method

A communication graph $G = (V, E)$ is defined to capture the spatial and temporal relationships among the drones, where the edges represent reliable signal links:

$$\begin{aligned} \min_{\mathbf{L}, \mathbf{S}} \sum_{i,n} p_\delta(W_{in}(M_{in} - L_{in} - S_{in})) + \lambda \|\mathbf{L}\|_* \\ + \frac{\gamma}{2} \text{tr}(\mathbf{L}^T \mathbf{L}_g \mathbf{L}) + \tau \sum_{i=1}^D \|\mathbf{D}_t \mathbf{L}_{i,:}\|_1 + \mu \|\mathbf{S}\|_1 \end{aligned} \quad (12)$$

In (12), p_δ is the Huber loss (with parameter $\delta = 1.345$ in our simulation), which ensures robustness to outliers, $\|\mathbf{L}\|_*$ promotes a globally low-rank structure among drone clocks, $\text{tr}(\mathbf{L}^T \mathbf{L}_g \mathbf{L})$ enforces spatial smoothness across connected drones, $\|\mathbf{D}_t \mathbf{L}_{i,:}\|_1$ penalizes abrupt temporal fluctuations, and $\|\mathbf{S}\|_1$ captures sparse disturbances caused by jamming or signal drops. This formulation exploits both temporal consistency and inter-drone correlations, yielding resilient and smooth synchronization even in dynamic and interference-prone aerial networks.

In (12), four hyperparameters, λ, γ, τ and μ , are employed to control different regularization strengths. λ is employed to regulate the low-rank nuclear norm, γ weights the graph Laplacian smoothness, τ is used to obtain temporal fused-lasso constraints, and μ governs the Huber-based robustness to outliers. These parameters were selected based on a validation search to ensure fairness; the same strategy was applied to Soft Impute. For the results presented, the key hyperparameters

were set to $\gamma = 0.05$, $\tau = 0.02$, and the nuclear norm penalty $\lambda \approx 4.68$ (which corresponds to $0.1s_1$ where s_1 is the largest singular value of the observation matrix).

The Graph Laplacian $\mathbf{L}_g$ is derived from communication connectivity among drones. Each drone is represented as a node in the graph $G = (V, E)$, and an edge (i, j) exists if drones i and j are able to reliably exchange timestamp messages within the observation window. In the proposed work, we construct a k-nearest neighbor (k-nn) graph where the edge weight (similarity) between two drones is determined using the Gaussian kernel of their relative communication quality, which is based on the distance between the drones. The weighted adjacency matrix $\mathbf{W}_g$ is used to calculate $\mathbf{L}_g = \mathbf{D}_g - \mathbf{W}_g$, where $\mathbf{D}_g$ represents the degree matrix.

V. PERFORMANCE EVALUATION

The proposed synchronization framework was evaluated on simulated multi-drone networks subject to communication disturbances, clock drifts, and offset variations.

We evaluated the framework on a simulated network of $D = 25$ drones over a 60-second period, sampled at 1 Hz, resulting in $N = 61$ time samples. The intrinsic clock parameters for each drone were generated randomly: initial offsets were drawn from a Gaussian distribution $\mathcal{N}(0, 0.5^2)$ (i.e., std. dev. of 0.5 s), and clock drifts were drawn from $\mathcal{N}(0, 0.01^2)$ (i.e., std. dev. of 0.01 s/s). To simulate a harsh environment, we introduced two types of impairments: (i) large-magnitude noise, affecting 3 “Jammed” drones (additive noise $\sim 5.0 \cdot \mathcal{N}(0, 1)$) and 2 “Low-signal” drones (additive noise $\sim 2.0 \cdot \mathcal{N}(0, 1)$); and (ii) data loss, where 2 “No-signal” drones had all measurements removed. four hyperparameters, λ, γ, τ and μ , are employed to control different regularization strengths. For the results presented, the key hyperparameters were set to $\gamma = 0.05$, $\tau = 0.02$, and the nuclear norm penalty $\lambda \approx 4.68$ (which corresponds to $0.1s_1$ where s_1 is the largest singular value of the observation matrix).

A random observation mask was applied to the remaining data, ensuring a minimum of 30% of total observations were preserved across the network.

Each method — Baseline, Fill+SVD, Soft Impute, Go-Decomposition, and GR-RLR — was evaluated for its ability to restore consistent timing across the fleet. The synchronization error for drone i is given by

$$e_i = \frac{1}{|\Omega_i|} \sum_{n \in \Omega_i} (\hat{y}_i[n] - t_n)^2, \quad (13)$$

where $\Omega_i = \{n, W_{in} = 1\}$ denotes the set of valid observations for drone i . The median RMSE and interquartile range across all drones summarize the aggregate performance of the proposed framework. For each environmental class C (jammed, low-signal, no signal), the absolute and percentage improvements of method m relative to the baseline are given by

$$\Delta_{\text{abs}}(m) = \text{median}_{i \in C} (e_i^{\text{Base}} - e_i^m), \quad (14)$$

and

$$\Delta_{\text{per}}(m) = \text{median}_{i \in C} \left(100 \cdot \frac{e_i^{\text{Base}} - e_i^m}{e_i^{\text{Base}}} \right). \quad (15)$$

The computational complexity of each method is $O(DN)$, $O(DN \min(D, N))$, and $O(DNr + K)$, for Baseline, matrix-completion, and GoDec variants, respectively. The GR-RLR model extends this to $O(DN \min(D, N)) + |E|N$, accounting for graph-based regularization. The framework is scalable and robust, making it well-suited for medium- to large-sized drone fleets that require synchronization across multiple locations.

VI. RESULTS AND DISCUSSION

The observed drone clock data are modeled according to the unified matrix formulation following (3). The goal of each synchronization method is to recover $\mathbf{L}$ and suppress $\mathbf{S}$ and $\mathbf{N}$ by minimizing the reconstruction loss in (13)-(15).

The aggregate performance of all five methods is summarized in Fig. 1 (Median RMSE with IQR) and Table II (precise values). The baseline affine detrending derived from (4) and (5) shows the highest error and widest IQR, indicating its inability to model correlated clock drifts or handle interference. In contrast, all decomposition-based methods, which leverage a shared low-rank structure, achieve significant improvements. As shown in Table II, the median RMSE improved from 0.5978 s for the baseline to 0.4458 s for Fill+SVD, Soft Impute, and GR-RLR, corresponding to a 25.4% reduction in synchronization error. The IQR was also reduced by over 46%, indicating a more consistent fleet. As shown in Table II, the median RMSE improved significantly from 0.5978 s (Baseline) to 0.4458 s for the decomposition-based methods. Notably, while GoDec achieves a marginally smaller IQR (0.5352) compared to GR-RLR (0.5371), this statistic warrants careful interpretation. GoDec's strict fixed-rank constraint enforces global uniformity, which reduces variance but may suppress valid local variations (leading to "over-smoothing"). In contrast, GR-RLR explicitly models adverse conditions through Huber loss and graph regularization. As seen in the RMSE Distribution (Fig. 2), GR-RLR achieves the most compact distribution in terms of outlier suppression, effectively rejecting the high-magnitude errors caused by jamming that GoDec fails to fully eliminate. Thus, GR-RLR offers the best balance between accuracy and operational safety. GR-RLR introduces three key innovations not present in Fill+SVD, Soft-Impute, or GoDec:

- 1) **Graph-based spatial regularization:** GR-RLR explicitly models correlations between neighboring drones via a Laplacian penalty. In contrast, Fill+SVD, Soft-Impute, and GoDec treat each drone independently.
- 2) **Temporal total variation:** GR-RLR enforces piecewise smoothness of clock errors over time, capturing gradual drift while preserving abrupt changes. Other methods lack any temporal constraints.
- 3) **Huber loss for robustness:** GR-RLR employs an adaptive Huber loss to down-weight large outliers, making it resilient to jamming and signal loss.

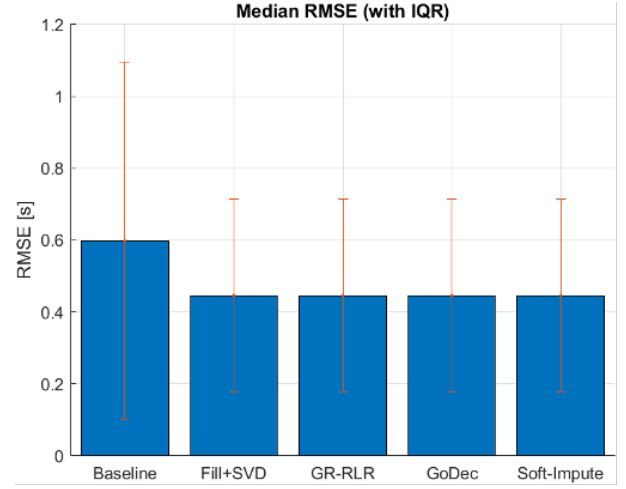


Fig. 1. Median RMSE with IQR

These features make GR-RLR more suitable for real-world scenarios with structured interference and non-uniform sensing quality. Its higher computational complexity is justified for applications where robustness under challenging conditions is critical. Table I summarizes the comparison.

TABLE I
FEATURE COMPARISON OF MATRIX COMPLETION METHODS

Feature	Fill+SVD	Soft-Impute	GoDec	GR-RLR
Spatial regularization	No	No	No	Yes
Temporal smoothness (TV)	No	No	No	Yes
Huber loss (robust)	No	No	No	Yes
Sparse corruption separation	Yes	Yes	Yes	Yes

Therefore, despite a negligible difference in IQR, GR-RLR provides superior robustness while maintaining the same median accuracy, thereby justifying its selection as the preferred method, as shown in Table III. While the median accuracy of the advanced decomposition methods is comparable, Fig. 2 (RMSE Distribution) reveals a critical difference in robustness. The baseline method exhibits a wide error dispersion and significantly high-magnitude outliers (circles) caused by jammed links. While all decomposition methods narrow this spread, the GR-RLR model achieves the most compact distribution, with the smallest interquartile range (IQR) and a clear suppression of all high-magnitude outliers. This demonstrates the superior stability of GR-RLR, which is a direct result of the Huber penalty and the spatio-temporal regularizers in (12) working together to reject extreme errors.

Across all reconstruction methods, while the aggregate median accuracy is similar, the proposed GR-RLR is demonstrably the most resilient and stable method under interference.

Fig. 3 links these findings to environmental classes. All methods under "Jammed" and "Low signal" conditions recover the clean manifold, lowering the median RMSE by $\approx 4s$ and $\approx 1s$, respectively, which is consistent with the accurate estimation of $\mathbf{L}$ and suppression of $\mathbf{S}$ described in (3). GR-RLR delivered the most stable recovery, indicating that the

TABLE II
SYNCHRONIZATION ACCURACY BETWEEN BASELINE METHOD AND
LOW-RANK DECOMPOSITION RECONSTRUCTION METHODS

Method	Median RMSE	IQR
Baseline method	0.5978	0.9955
Fill+SVD	0.4458	0.5371
Soft Impute	0.4458	0.5371
GO-Decomposition	0.4458	0.5352
GR-RLR	0.4458	0.5371

TABLE III
COMPARISON OF DIFFERENT METHODS

Method	Low-Rank	Spikes	Graph	Smooths	Time
Baseline	×	×	×	×	×
Fill+SVD	✓	×	×	×	×
Soft-Impute	✓	×	×	×	×
GoDec	✓	✓	×	×	×
GR-RLR	✓	✓	✓	✓	✓

low-rank structure, graph Laplacian regularization, and Huber-based loss work together to achieve resilient synchronization that is more robust and reliable in a dynamic multi-drone network.

Fig. 4 shows how the percentage recovery performance is derived from (14) and (15). All approaches attained median recovery of over 80% in the "Jammed" environments class, validating the model's capacity to suppress significant-magnitude sparse disturbances in (3). In the "Low Signal" environmental class, approximately 60% recovery is achieved, indicating that low-rank $\mathbf{L}$ effectively reconstructs partially corrupted temporal patterns. As predicted by (12), GR-RLR performs slightly better than other approaches by leveraging the spatial Laplacian $\mathbf{L}_g\mathbf{L}$ and temporal derivative $\mathbf{D}_t\mathbf{L}$, enhancing recovery, which is more consistent across neighboring drones. In the "No Signal" environmental class, negligible recovery is consistent with the absence of a valid timestamp.

Fig. 5 shows the Soft-Impute validation curve, showing

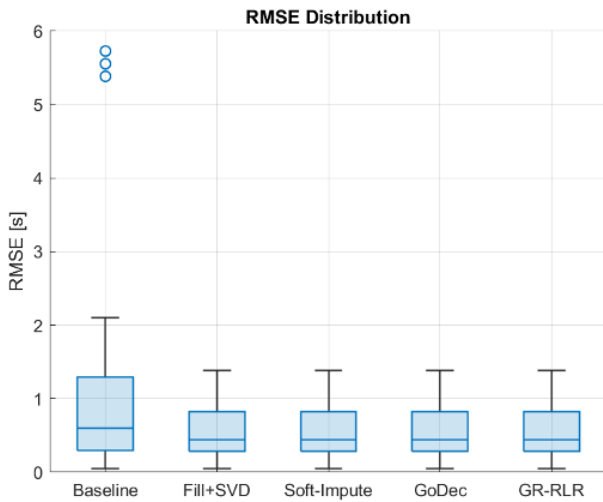


Fig. 2. RMSE distribution

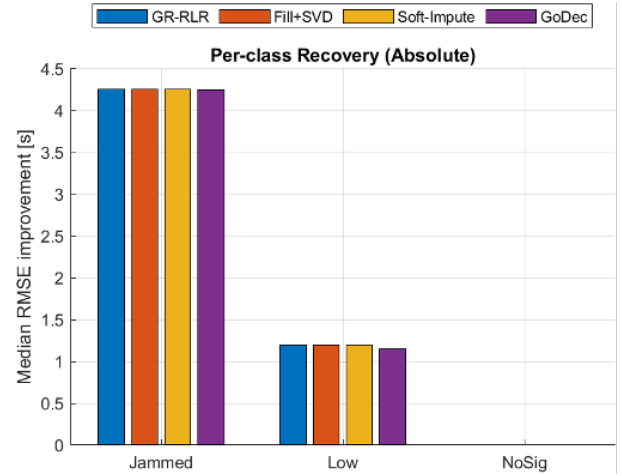


Fig. 3. Per Class recovery (Absolute)

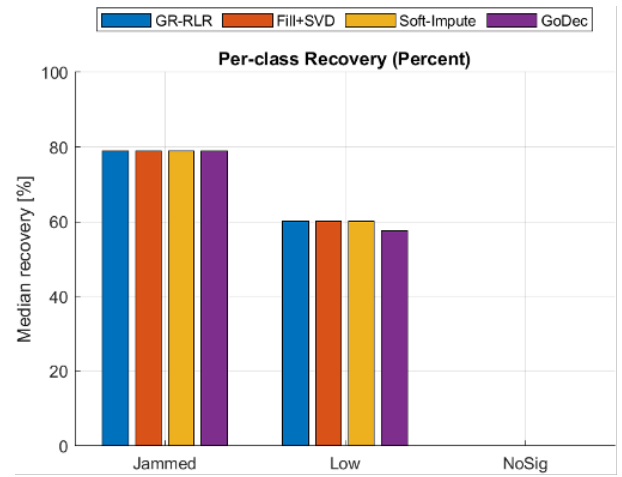


Fig. 4. per-class recovery (Percent)

how the validation RMSE changes with the nuclear-norm regularization parameter λ in (9)–(10). The curve describes a shallow minimum at $\lambda \approx 1 - 2$, illustrating the trade-off between fidelity and overregularization. This empirical behavior validates the role of $\|\mathbf{L}\|_*$ in facilitating a compact yet adaptable temporal representation.

Fig. 6 illustrates a heatmap of RMSE per drone and approach, illustrating the uniformity of spatial synchronization. For the decomposition-based models, high RMSE bands in the baseline are significantly reduced, especially in GR-RLR, resulting in uniform low error across all drones. This validates the inclusion of the Laplacian regularizer $\text{tr}(\mathbf{L}^T\mathbf{L}_g\mathbf{L})$ in (12), which enforces coherent corrections across spatially correlated nodes.

The results reveal the parameter stability, environmental robustness, and spatial uniformity achieved through the unified low-rank–sparse–graph formulation. Together, these findings demonstrate that GR-RLR provides a scalable and resilient solution for multi-drone clock synchronization under dynamic, interference-dominated conditions.

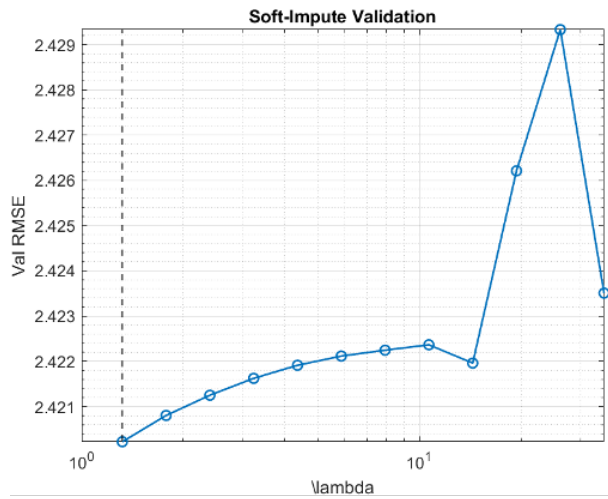


Fig. 5. Soft Impute validation

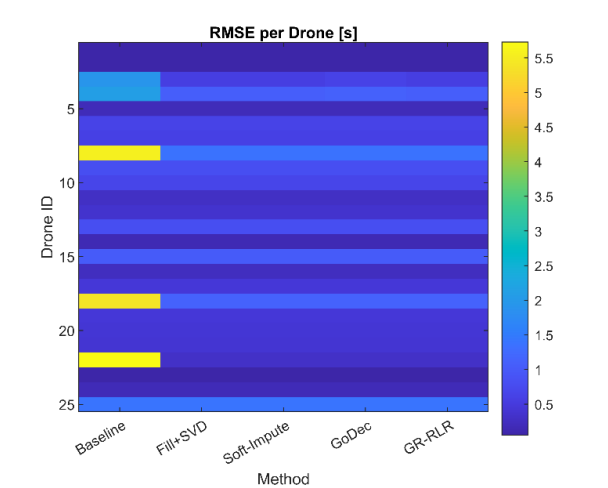


Fig. 6. Heat MAP of RMSE per drone

VII. CONCLUSION

This paper developed a comprehensive framework for multi-drone clock synchronization in dynamic, interference-prone environments. By comparing five distinct strategies, we demonstrated the limitations of classical consensus and standard matrix completion in handling sparse corruptions.

The proposed Graph-Regularized Robust Low-Rank (GR-RLR) model emerged as the most resilient solution. By explicitly integrating Huber data fidelity with spatio-temporal graph priors, GR-RLR effectively separates clean clock drifts from sparse jamming interference. Although standard methods like GoDec improve central accuracy, GR-RLR distinguishes itself by maintaining superior stability and suppressing extreme outliers in low-signal scenarios.

Future work will extend this evaluation to include extensive multi-seed Monte Carlo simulations to rigorously quantify tail-risk metrics. Furthermore, we aim to develop hybrid batch-online algorithms to bridge the gap between static decom-

position and real-time filtering, paving the way for fully autonomous, communication-resilient drone swarms.

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